

REBOLY, Russian Federation

WMO: 226020

Lat: 63.833N

Lon: 30.817E

Elev: 182

StdP: 99.16

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.1	(c) -26.9	(d) -33.1	(e) 0.2	(f) -30.0	(g) -29.8	(h) 0.2	(i) -26.9	(j) 6.4	(k) -3.7	(l) 6.2	(m) -3.8	(n) 0.3	(o) 250	(p) 0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	25.8	17.7	23.8	16.7	22.0	15.7	19.0	23.2	17.8	22.3	16.7	20.8	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	12.7	20.9	16.1	11.7	19.7	14.9	10.8	18.6	54.6	23.1	50.8	22.4	47.4	20.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.2	5.4	5.0	-32.7	28.4	3.7	2.5	-35.4	30.2	-37.5	31.7	-39.6	33.1	-42.3	35.0
			-32.8	20.3	3.4	2.3	-35.3	21.9	-37.2	23.3	-39.1	24.6	-41.6	26.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-10.7	-10.4	-5.6	0.8	7.5	13.0	16.3	14.0	9.0	2.8	-3.0	-7.2		
	DBStd	10.70	7.76	8.16	5.57	4.54	4.87	3.97	3.45	3.28	3.42	4.00	5.14	7.18		
	HDD10.0	3307	642	571	483	277	108	14	1	4	58	224	391	533		
	HDD18.3	5894	901	804	742	527	338	168	83	141	279	481	641	791		
	CDD10.0	487	0	0	0	1	29	103	196	128	29	1	0	0		
	CDD18.3	32	0	0	0	0	1	7	19	5	0	0	0	0		
	CDH23.3	209	0	0	0	0	11	45	120	33	0	0	0	0		
	CDH26.7	26	0	0	0	0	1	6	16	3	0	0	0	0		
(14) Wind	WSAvg	2.2	2.3	2.2	2.3	2.1	2.3	2.3	2.2	2.0	2.3	2.4	2.4	2.3		
(15) Precipitation	PrecAvg	596	34	30	31	35	46	59	70	75	65	58	49	41		
	PrecMax	806	67	58	73	99	119	141	130	154	145	130	123	80		
	PrecMin	397	8	3	2	4	1	20	14	12	16	14	16	11		
	PrecStd	95	14	14	17	21	26	27	29	34	28	27	22	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.5	3.0	8.0	16.4	25.0	27.3	28.9	26.8	20.2	12.8	7.4	3.4		
		MCWB	1.7	1.4	3.6	9.1	14.8	17.6	19.2	18.2	14.4	10.9	6.5	3.0		
	2%	DB	1.0	1.5	4.7	11.9	21.4	24.3	26.5	23.9	17.6	10.6	5.9	2.1		
		MCWB	0.2	0.6	1.3	6.5	13.0	16.2	18.3	17.1	13.3	9.4	5.0	1.4		
	5%	DB	0.3	0.6	2.7	9.4	18.3	22.1	24.6	21.9	15.7	9.4	4.3	1.2		
		MCWB	-0.3	0.0	0.4	4.7	11.2	15.2	17.5	16.1	12.4	8.3	3.6	0.5		
	10%	DB	-0.6	-0.2	1.3	7.3	15.7	20.1	22.6	19.9	14.2	8.2	2.8	0.5		
		MCWB	-1.1	-0.7	-0.3	3.5	9.8	14.1	16.6	15.3	11.8	7.0	2.2	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	1.8	3.7	9.7	16.2	19.6	21.2	19.7	15.6	11.1	7.1	3.1		
		MCDB	2.3	2.4	7.5	14.7	22.6	23.2	26.2	24.3	18.5	12.0	7.3	3.2		
	2%	WB	0.5	0.7	1.9	7.3	13.9	17.5	19.5	18.0	14.2	9.7	5.2	1.7		
		MCDB	0.9	1.3	4.0	11.2	19.3	22.6	24.1	22.2	16.4	10.4	5.7	2.0		
	5%	WB	-0.2	0.1	0.8	5.3	12.2	16.2	18.4	16.8	13.1	8.5	3.7	0.7		
		MCDB	0.2	0.5	2.2	8.4	17.1	21.0	22.9	20.6	14.9	9.3	4.1	1.0		
	10%	WB	-1.1	-0.8	0.0	3.9	10.4	14.6	17.4	15.8	12.2	7.1	2.3	-0.1		
		MCDB	-0.6	-0.3	1.2	6.8	14.8	18.7	21.5	19.3	13.9	7.9	2.8	0.4		
(35) Mean Daily Temperature Range	MDBR		6.3	7.2	8.4	8.6	8.5	8.0	7.8	7.4	6.0	3.9	4.1	5.3		
	5% DB	MCDBR	3.7	5.0	8.3	11.7	13.6	10.9	10.6	10.7	8.5	4.8	3.9	3.9		
		MCWBR	3.8	4.9	6.2	7.1	6.9	4.9	4.5	4.8	5.1	3.9	3.8	3.7		
	5% WB	MCDBR	3.5	4.8	6.6	10.5	11.7	10.0	9.2	9.1	6.6	4.6	3.7	3.7		
		MCWBR	3.7	4.7	5.3	6.7	6.6	5.1	4.5	4.9	4.7	4.2	3.8	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.217	0.279	0.307	0.342	0.348	0.344	0.371	0.349	0.326	0.304	N/A	0.205		
	taud		1.961	2.095	2.142	2.254	2.445	2.485	2.427	2.467	2.516	2.385	N/A	1.953		
	Ebn at Noon		356	624	771	831	865	876	836	824	772	618	N/A	153		
	Edh at Noon		29	65	94	106	97	95	98	86	67	50	N/A	13		
(44) All-Sky Solar Radiation	RadAvg		0.16	0.72	2.18	3.71	4.68	5.16	4.86	3.65	2.03	0.73	0.17	0.06		
	RadStd		0.02	0.15	0.24	0.35	0.56	0.35	0.52	0.37	0.22	0.11	0.03	0.01		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	DBAvg	+0.59	N/A	N/A	N/A	N/A	N/A	-192	N/A

Nomenclature: See separate page